EDUHUB / ENGINEERING DOCUMENTATION

# TECHNOLOGY STACK

EduHub | Online Learning Platform

| | |
|---|---|
| 01 PRESENTATION | ASP.NET Core MVC / Razor / Bootstrap 5 |
| 02 APPLICATION | C# / Services / Identity / Design Patterns |
| 03 DATA & INTEGRATIONS | EF Core / SQL / Azure / Stripe / Media |

Technology selection, integration, lifecycle and tooling.

Source requirements are distinguished from engineering recommendations.

Prepared from the EduHub project brief and established design baseline.

# 01 | Technology direction

Decision baseline, document scope and evidence classification.

## 01 EXECUTIVE POSITION

EduHub is a **server-rendered ASP.NET Core MVC learning platform** delivered initially as a **three-tier modular monolith**. The required browser libraries, Entity Framework Core data access, Stripe payment integration and Azure media hosting remain part of the approved source baseline. [S1]

**CORE ARCHITECTURE DECISION**

Preserve **EduHub.Web** (presentation), **EduHub.Services** (application logic) and **EduHub.Data** (persistence), deployed initially as one application plus managed external services. This matches the Software Architecture deliverable without introducing microservices.

## 02 HOW TO READ THIS DOCUMENT

| STATUS | MEANING | DECISION RULE |
|---|---|---|
| REQUIRED | Explicit in the five-page EduHub brief. | Do not replace without course/supervisor approval. |
| DERIVED | Necessary to implement a required workflow or already established in prior design. | Confirm implementation detail with team. |
| RECOMMENDED | Engineering choice proposed here (version, package, tool or Azure feature). | Approve in the decision register; can be substituted. |

## 03 SCOPE BOUNDARY

This document answers **which technologies are used, where they run, why they are suitable, how they integrate, and what constraints apply**. It does not replace the dedicated Component Diagram, Deployment Diagram, API contract, or Testing Plan.

**SOURCE [S1]** Five-page **EduHub - Online Learning Platform** brief (project-specific authority).

**BASELINE** Prior Software Architecture, ERD, Logical/Physical Schema, use cases, DFD and UML artifacts (cross-document consistency).

# 02 | Complete technology inventory

Specified tooling and proposed implementation choices.

## 01 SOURCE-MANDATED PLATFORM & LIBRARIES

| TECHNOLOGY | ROLE IN EDUHUB | STATUS |
|---|---|---|
| ASP.NET Core MVC | MVC controllers, Razor views and request routing. | REQUIRED |
| Entity Framework Core | ORM for application persistence. | REQUIRED |
| ASP.NET Core Identity | Login, users, claims and Student/Instructor/Admin roles. | REQUIRED |
| Azure Blob Storage | Course videos, thumbnails and generated media. | REQUIRED |
| Stripe | Paid-course checkout and verified payment events. | REQUIRED |
| iTextSharp | PDF certificate generation (legacy product risk). | REQUIRED |
| Video.js | Student video player. | REQUIRED |
| Progress.js | Learner progress UI; exact package to confirm. | REQUIRED |
| Bootstrap 5 | Responsive UI components and layout. | REQUIRED |
| Chart.js | Sales, earnings and engagement dashboards. | REQUIRED |
| Azure App Service | Host the application on Microsoft Azure. | REQUIRED |
| Azure CDN | Deliver video through CDN; classic SKU review needed. | REQUIRED |

## 02 RECOMMENDED IMPLEMENTATION COMPLETION

**.NET 10 LTS / C#**, **Azure SQL Database (SQL Server)**, Azure managed identities, Git-based CI/CD, automated tests and monitoring are recommended configuration choices; the original brief does not mandate their exact versions or products. [R1-R4, R8]

# 03 | Layered stack map

Logical dependency and responsibility boundaries.

## 01 APPLICATION LAYERS AND BACKING SERVICES

### CLIENT EXPERIENCE

Browser | Razor HTML | Bootstrap 5 | Video.js | Progress.js | Chart.js



### EDUHUB.WEB

ASP.NET Core MVC | Controllers | Views | Authorization | Model validation



### EDUHUB.SERVICES

Courses | Learning | Commerce | Certificates | Admin | Interfaces



### EDUHUB.DATA

EF Core | Identity stores | Repositories | Unit of Work | Migrations

## 02 EXTERNAL ADAPTERS (NOT EXTRA APPLICATION TIERS)

| STORAGE | PAYMENT | PDF / DISTRIBUTION |
|---|---|---|
| Azure SQL + Blob Storage | Stripe Checkout and signed webhooks | iTextSharp certificate adapter; Azure CDN or approved Front Door SKU |

**Dependency rule:** MVC controllers invoke application services; services coordinate domain operations through interfaces; EF Core repositories implement persistence; Stripe and Azure concerns remain behind adapters. Client code never receives database credentials or secret API keys.

# 04 | Presentation technology

What runs in the browser and in the ASP.NET Core MVC presentation project.

## 01 WEB EXPERIENCE MATRIX

| CAPABILITY | TECHNOLOGY | INTEGRATION NOTE |
|---|---|---|
| Pages and routing | ASP.NET Core MVC + Razor | Server-rendered catalog, dashboards, profiles, course pages. |
| Responsive design | Bootstrap 5 + CSS | Accessible forms, cards, navigation and device layouts. |
| Video playback | Video.js | Player receives authorized media URL after entitlement check. |
| Progress indicator | Progress.js | Displays lesson/course completion; server remains authoritative. |
| Dashboards | Chart.js | Consumes aggregated, permission-scoped reporting data. |
| Interactive builder | JavaScript + HTML drag/drop | Custom client script; persist order through validated MVC endpoints. |

## 02 BROWSER-TO-SERVER CONTRACT

- **Views:** Razor `.cshtml` templates; keep complex business logic in service layer, not Razor or controllers. [R2]
- **Commands:** antiforgery-protected form POSTs or authenticated JSON requests for editing, progress, quizzes and cart updates.
- **Streaming:** video bytes come from protected storage/distribution endpoint; do not proxy full media through MVC unless required.
- **Charts:** aggregation takes place server-side; never expose other students' personal records.

> **IMPORTANT PACKAGE CLARIFICATION**
>
> The brief spells **"Progress. js"** but does not identify a package repository/version. Confirm the exact library before locking dependencies; **Bootstrap progress** is a fallback only if replacing it is approved. [S1, R10]

# 05 | Application & design patterns

C# services and architectural patterns required by the brief.

## 01 BACKEND TECHNOLOGY ASSIGNMENTS

| ITEM | PLACEMENT | ENGINEERING RATIONALE |
|---|---|---|
| C# / .NET 10 LTS | Web, Services, Data | Suggested supported runtime and language for the stack. [R1] |
| ASP.NET Core MVC | EduHub.Web | Routes, binding, validation, cookies, antiforgery, Razor. [R2] |
| Dependency Injection | Composition root: Web | Register services and adapters by interfaces. |
| Repository | EduHub.Data | Abstract data operations; avoid repository methods that duplicate all EF APIs. |
| Unit of Work | EduHub.Data | Use shared EF Core DbContext transaction/SaveChanges for coordinated writes. |
| Strategy | EduHub.Services | Coupon, fixed and percentage discount / pricing rules. |
| Observer / events | EduHub.Services | After durable progress updates, publish notifications or completion actions. |

## 02 SOLUTION STRUCTURE (REQUIRED PROJECTS)

```
EduHub.sln
  EduHub.Web       // MVC, views, request pipeline, DI
  EduHub.Services  // business rules, DTOs, interfaces
  EduHub.Data      // DbContext, repositories, migrations
  EduHub.Tests     // recommended independent test project
```

**Architectural guardrail:** keep one deployable MVC application initially. Use internal feature modules and interfaces instead of independently deployed microservices. External media processing can be attached later if justified.

# 06 | Database and identity

Persistence choices tied to the existing ERD and logical/physical schema.

## 01 DATA PLATFORM & ROLE BOUNDARIES

| TECH | DECISION | WHY IT FITS |
|---|---|---|
| Entity Framework Core | Required ORM | Mappings, migrations, LINQ queries and transactions. |
| SQL Server / Azure SQL | Recommended relational engine | Matches prior T-SQL physical schema and 38 modeled entities. |
| ASP.NET Core Identity | Required user platform | Three roles, credential operations, claims/tokens. [R12] |
| EF Core SQL Server provider | Recommended NuGet package | Microsoft.EntityFrameworkCore.SqlServer. [R3] |

## 02 PERSISTENCE RULES

**ENTITY MODEL** Use the **38-entity logical design** as the model of record; keep the three Identity entities mapped to framework ownership and extend only as needed.

**TRANSACTIONS** Persist course edits, quiz attempt grades, enrollment creation and order settlement with atomic boundaries where database consistency requires them.

**INTEGRITY** Enforce unique enrollment, one-time payment event handling, lesson order, primary/foreign keys and status constraints in the database *and* services.

**PRIVACY** Credentials/password hashes stay in Identity storage; never store card numbers, CVC or unnecessary payment PII in application tables.

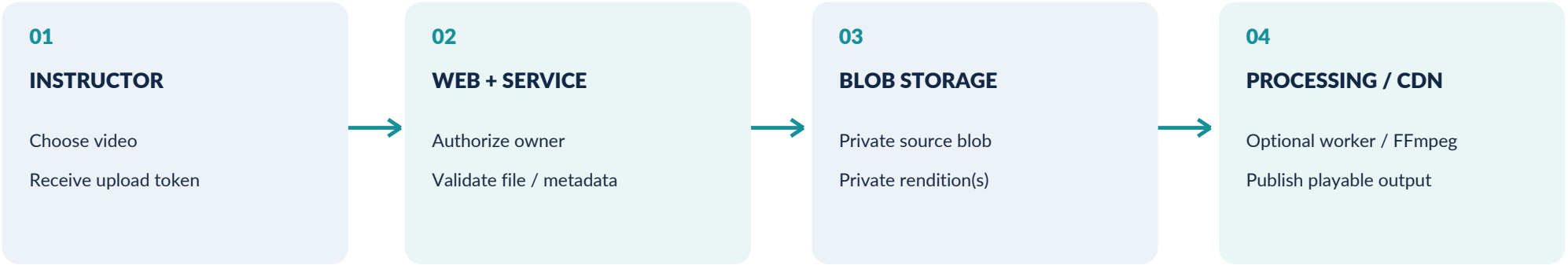
## 03 SOURCE-OF-TRUTH DISTINCTIONS

> **SUPPORTED ARCHITECTURAL CHOICE**
>
> The brief requires **EF Core**, but does **not** explicitly name SQL Server. This database was proposed in the earlier schema document; adopting Azure SQL is a logical continuation, not a newly mandated requirement. [S1, R3]

# 07 | Video pipeline & protected playback

Upload, transform, authorize and deliver; recommended processing implementation is identified separately.

## 01 MEDIA PIPELINE (VECTOR TECHNOLOGY VIEW)




## 02 PLAYBACK AUTHORIZATION & OWNERSHIP

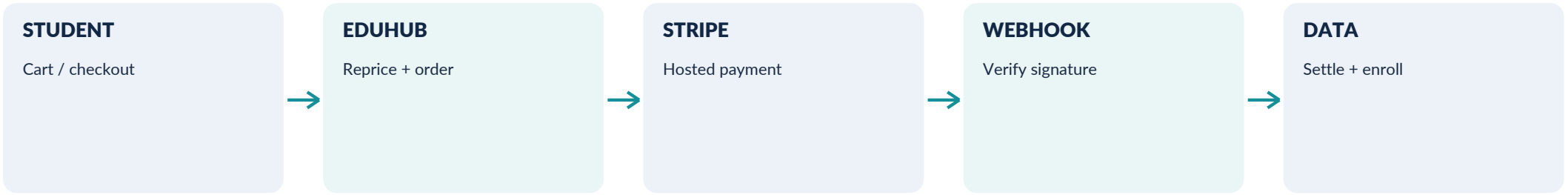
| STEP | TECHNOLOGY | ACCESS CONTROL |
|---|---|---|
| 1. Student requests lesson | MVC endpoint + Services | Verify enrollment, lesson availability and policy. |
| 2. Issue scoped media access | Azure Blob SAS or compatible signed edge token | Expire quickly; no permanent public blob URLs. [R4] |
| 3. Load video | Video.js + CDN/edge | Browser fetches permitted media only. |
| 4. Record completion | MVC command + EF Core | Server validates genuine lesson and enrollment. |

**Processing gap:** the brief requires compression and thumbnail creation but names no processor. A background worker / Azure Function plus FFmpeg is one **recommended** implementation, subject to operational and codec licensing review. **CDN note:** private video caching must be tested with the actual signed-URL scheme.

# 08 | Stripe commerce integration

Payment authority comes from verified webhook fulfillment, not browser success redirects.

## 01 PURCHASE AND FULFILLMENT PATH




## 02 INTEGRATION BOUNDARIES AND FAILURE RULES

| ELEMENT | REQUIRED / RECOMMENDED CONTROL |
|---|---|
| Stripe SDK / Checkout | Required provider; Stripe-hosted checkout recommended to minimize PCI exposure. |
| Webhook endpoint | Verify Stripe signature using raw request body; acknowledge quickly. [R5] |
| Idempotency | Unique Stripe event ID; database transaction must not create duplicate enrollments. |
| Redirect page | Informational only; never a payment-settlement authority. |
| Coupon Strategy | Recalculate final amount in trusted service before Stripe session creation. |
| Free course path | Enrollment can occur without a Stripe payment when eligible. |

# 09 | Learning, analytics & certificates

Technology decisions around learner progress, assessments and proof of completion.

## 01 FEATURE-TO-COMPONENT MAPPING

| FEATURE | TECHNOLOGY | DELIVERY BEHAVIOR |
|---|---|---|
| Lesson viewing | Video.js + ASP.NET MVC | Protected playback and explicit lesson completion. |
| Progress tracking | EF Core + Progress.js | Server persists completion; UI renders % from authoritative data. |
| Quiz grading | C# services + EF Core | Question validation, attempts, pass threshold and explanations. |
| Notifications | Service event/Observer | Emit notification after progress/course transactions commit. |
| Certificate PDF | iTextSharp (required) | Create personalized, uniquely verifiable PDF. |
| Analytics | Chart.js + reporting service | Course sales, earnings, completions and engagement. |

## 02 CERTIFICATE TECHNOLOGY DECISION

The project brief explicitly names **iTextSharp**. The vendor identifies iTextSharp/iText 5 as **end-of-life** and recommends a newer iText Core implementation for new projects. Both involve **AGPL/commercial licensing** considerations. Retain iTextSharp for strict brief compliance or obtain approval to adopt a supported alternative; document the license decision. [S1, R6]

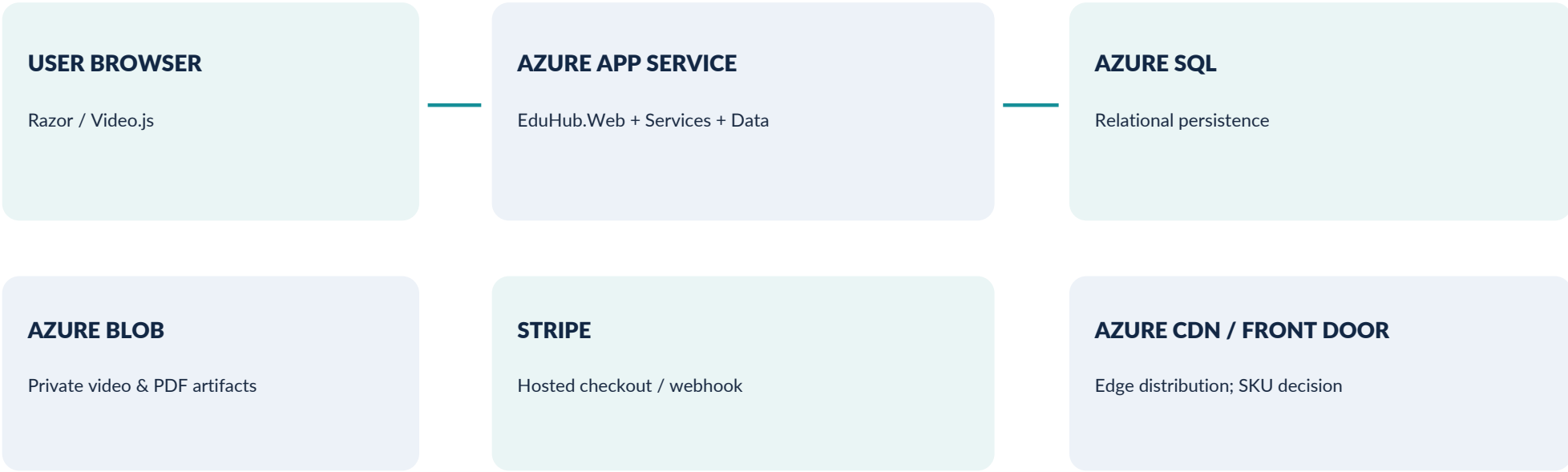
## 03 ACCURACY AND EVENT GUARANTEES

- **Completion eligibility:** derive certificate eligibility from persisted lesson and required quiz results; do not rely on a JavaScript progress bar.
- **Retry protection:** prevent two certificates for the same enrollment unless deliberate reissuance is modeled.
- **Analytics:** aggregates are advisory dashboards, not an alternative system of record for purchases or progress.

# 10 | Azure hosting topology

Proposed operational environment; Azure is required, the precise services and SKUs need approval.

## 01 LOGICAL DEPLOYMENT DIAGRAM




## 02 OPERATIONAL SERVICES (OPTIONAL CHOICES)

| BASELINE | SCALING / HARDENING RECOMMENDATION |
|---|---|
| App Service + SQL + private Blob | Managed identity / Key Vault, telemetry, backups and environment separation. |
| Original brief: Azure CDN | Microsoft retires CDN Standard from Microsoft (classic) in Sep 2027; review Front Door Standard/Premium instead. [R7] |
| Initial single application | Add caching, background worker or queue only when requirements and load justify complexity. |

# 11 | Security and operational controls

Controls tied directly to the selected technologies and external integrations.

## 01 SECURITY-CONTROL IMPLEMENTATION MATRIX

| AREA | STACK OWNER | MINIMUM DESIGN CONTROL |
|---|---|---|
| Authentication | ASP.NET Core Identity | Hashed passwords, secure cookies, confirmation/reset flow. |
| Authorization | MVC + Services | Role policies and per-course / per-enrollment ownership checks. |
| Input protection | MVC / Razor | Server-side validation, antiforgery, output encoding. |
| Payments | Stripe adapter | Signed webhook validation, event deduplication, server-side total. |
| Video | Azure Blob/edge | Private containers and short-lived scoped playback access. |
| Secrets | App Service configuration | No keys in Git, browser, logs or compiled constants. |
| Database | EF Core + SQL | Parameterization, least-privilege accounts, safe migrations. |
| Auditability | Service events / logging | Trace course moderation, failed payments, certificate issue. |

## 02 RESILIENCE PRIORITIES

- Payment and certificate workflows use idempotent commands; database rollback does not undo an external Stripe action.
- Retries of transient Azure/SQL failures must be bounded and logged; processing cannot silently duplicate effects.
- Store only necessary learner metrics; limit dashboard access to legitimate roles and applicable ownership.

> **DO NOT CONFUSE CLOUD HOSTING WITH SECURITY**
>
> Moving to Azure does not itself guarantee media authorization, secret rotation, payment reconciliation, accessibility, or data protection. These require application-level enforcement. [R4, R5]

# 12 | Tooling, quality & delivery

Supporting developer tools are recommendations, not explicit requirements in the project brief.

## 01 DEVELOPER WORKBENCH

| PURPOSE | RECOMMENDED TOOL | WHY |
|---|---|---|
| IDE | Visual Studio / VS Code + .NET SDK | Build, debug and publish all three projects. |
| Source control | Git + GitHub | PR review, branches and code history. |
| Dependency management | NuGet + npm / local asset pipeline | Track .NET and browser packages with lock policies. |
| Automated testing | xUnit + ASP.NET Core test host | Unit and end-to-end HTTP integration tests. |
| Browser testing | Playwright (optional) | Student enrollment, lesson and admin journeys. |
| CI/CD | GitHub Actions or Azure Pipelines | Restore, build, test, scan and deploy. |
| Telemetry | Application Insights / Azure Monitor | Errors, latency, external dependency health. |

## 02 RELEASE QUALITY GATES (PROPOSED)

**BUILD** `dotnet restore` and `dotnet build` succeed on the pinned SDK and dependency versions.

**TEST** Domain service tests plus integration checks for Identity, EF Core and verified Stripe webhooks.

**SECURITY** Audit NuGet/npm dependency advisories, configuration and user authorization paths.

**STAGING** Run migration and smoke tests against isolated test configuration before production publish.

# 13 | Compatibility and lifecycle

Choose supported, interoperable versions; do not equate the project brief with package-version approval.

## 01 VERSION AND SUPPORT POLICY

| COMPONENT | PROPOSED BASELINE | VERIFY BEFORE COMMIT |
|---|---|---|
| .NET SDK / Runtime | .NET 10 LTS | Microsoft support to Nov 2028; pin SDK via global.json. [R1] |
| ASP.NET Core MVC | .NET 10 line | Match target framework and hosted runtime. [R2] |
| EF Core + SQL provider | EF Core 10 family | Match major versions to .NET runtime; migrations tests. [R3] |
| ASP.NET Core Identity | .NET 10 line | Use supported Identity/EF Core integration. [R12] |
| Bootstrap 5 | Current supported 5.x | Validate build and progress-markup behavior. [R10] |
| Video.js | Selected supported release | Check HTML/Razor setup and codecs/browser support. [R9] |
| Chart.js | Pinned supported release | Check browser chart and accessibility support. |
| Progress.js | Version unknown in brief | Confirm exact library and maintenance status. |
| iTextSharp | Brief-mandated legacy | End-of-life and AGPL/commercial review. [R6] |
| Azure CDN | Brief-mandated brand | Avoid classic retiring SKU; sign off replacement. [R7] |

## 02 LICENSING AND SUPPLY-CHAIN GATE

Lock package versions after smoke-testing the chosen .NET runtime; inventory transitive dependencies, build third-party notices, check NuGet/npm vulnerability advisories, and seek supervisor approval before deviating from explicitly required libraries. Licensing is particularly important for PDF engines and any proposed media transcoder.

# 14 | Reference packages & service adapters

Concrete implementation candidates: not a final version-pinned dependency lockfile.

## 01 NUGET DEPENDENCY SHORTLIST

| PACKAGE / FEATURE | PROJECT | PURPOSE |
|---|---|---|
| Microsoft.AspNetCore.Identity.EntityFrameworkCore | Web / Data | Identity with EF stores; use framework-aligned version. |
| Microsoft.EntityFrameworkCore.SqlServer | Data | SQL Server and Azure SQL provider. |
| Microsoft.EntityFrameworkCore.Design | Data (design-time) | Migrations and DbContext tooling. |
| Azure.Storage.Blobs | Data / adapter | Private video upload, metadata and scoped SAS. |
| Stripe.net (official SDK) | Services adapter | Checkout session / payment operations. |
| iTextSharp package (legacy) | Services adapter | PDF certificate generation only if approved. |
| Microsoft.Extensions.* framework APIs | Web / Services | DI, logging, options and hosted services. |

## 02 FRONTEND ASSET SHORTLIST

| ASSET | LOCATION | RULE |
|---|---|---|
| Bootstrap 5 styles/scripts | EduHub.Web/wwwroot | Pin version, load local/bundled or approved CDN. |
| Video.js | Student lesson view | Do not bind to public blob URI. |
| Progress.js | Learning dashboard | Confirm package before integration. |
| Chart.js | Admin + instructor dashboards | Get aggregates from protected server endpoints. |

**Build hygiene:** the table specifies products and package names as candidates. Exact patch releases, package licenses, and supported target-framework combinations must be verified during dependency installation, not inferred from this PDF.

# 15 | Risk register and architecture decisions

Deliberate trade-offs requiring explicit team ownership.

## 01 DECISION/RISK MATRIX

| ID / RISK | IMPACT | MITIGATION / ACTION |
|---|---|---|
| D-01 iTextSharp legacy / AGPL | HIGH | Confirm required library or approve supported PDF engine and acceptable license. [R6] |
| D-02 Azure CDN classic retirement | HIGH | Approve a supported Azure Front Door Standard/Premium or current CDN SKU. [R7] |
| D-03 Progress.js ambiguous product | MEDIUM | Select exact package/version; test accessibility; approve fallback if necessary. |
| D-04 Media transcoding unspecified | MEDIUM | Choose worker/queue + processor only after codec, cost and storage review. |
| D-05 App Service and SQL cost | MEDIUM | Agree subscription, region, student-credit/budget limits and retention. |
| D-06 Stripe production enablement | MEDIUM | Use test-mode keys early; validate country, account and production eligibility. |
| D-07 Data schema drift | HIGH | Compare EF migrations with 38-entity approved logical model. |
| D-08 Broad stack upgrades | MEDIUM | Freeze compatible versions and run full regression tests before upgrading. |

## 02 SIGN-OFF POLICY

A deviation that replaces a required technology (especially iTextSharp, Progress.js or Azure CDN) must be recorded as **approved substitution**, not described retrospectively as if it were in the original brief. Once decisions D-01 through D-08 are signed off, update architecture, deployment and test deliverables accordingly.

# 16 | Recommended implementation order

Dependency-aware adoption rather than a list of unrelated products.

## 01 FOUR DELIVERY WAVES ALIGNED WITH THE PROJECT PLAN

| WAVE | STACK ELEMENTS | VERIFIABLE EXIT CRITERION |
|---|---|---|
| W1 - Foundation | ASP.NET Core MVC; three projects; EF Core; SQL; Identity; Bootstrap 5 | Builds, migrations apply, Student/Instructor/Admin authentication works. |
| W2 - Authoring | Blob Storage; secure upload; processing choice; lesson builder | Instructor can draft and publish approved protected lessons. |
| W3 - Learning / Commerce | Video.js, Progress.js, Stripe, quizzes, Chart.js basics | Entitlements, progress and payments reconcile with database records. |
| W4 - Governance / Release | iTextSharp/approved PDF; full Chart.js dashboards; Azure App Service; CDN/Front Door | Certificates verify, reports reconcile, secure hosted release passes tests. |

## 02 MINIMUM VIABLE TECHNICAL PROOF

- Demonstrate an instructor creating a course and uploading a private lesson.
- Demonstrate a student enrolling free or via a verified Stripe test webhook (including duplicate delivery).
- Demonstrate entitlement-gated playback, quiz completion and a verifiable certificate.
- Show authorization failures when one instructor/student accesses another actor's protected resources.

### WHAT NOT TO ADD BY DEFAULT

Do not introduce Kubernetes, a SPA rewrite, microservices, external event buses or a second database solely to look "enterprise". Adopt them only when the approved requirements create a demonstrable need.

# 17 | Source-to-stack traceability

Every required major technology maps to an explicit project capability.

## 01 TRACEABILITY REGISTER

| SOURCE REQUIREMENT [S1] | TECHNOLOGY / DESIGN | COVERAGE |
|---|---|---|
| NTier + Repo / UoW / Strategy / Observer / DI | C# three projects; EF Core; DI; app strategies/events | Architecture / services |
| Student / Instructor / Admin | ASP.NET Core Identity + authorization | Access control |
| Course creation / publication | ASP.NET Core MVC + Razor + EF Core | Content management |
| Media upload / compression / signed URLs | Azure Blob; processing adapter; protected access | Video lifecycle |
| Video playback / progress | Video.js + Progress.js | Student learning |
| Paid checkout / coupons | Stripe + strategy interface | Commerce |
| Quizzes / results | C# grading services + EF Core | Assessment |
| Certificate PDF / verification | iTextSharp + certificate service | Certification |
| Reports / admin analytics | Chart.js + reporting service | Reporting |
| Azure deployment / video CDN | Azure App Service + CDN/approved edge service | Operations |

## 02 CONSISTENCY NOTES

The technologies here do not alter the previously documented **three-layer deployment unit, 38 persistent entities, 48 use cases, 83 functional and 32 non-functional requirements**. The technology document is a selection reference; those earlier artifacts remain the authorities for their own scopes.

# 18 | Sources & technical references

Official vendor and platform documentation; accessed 05 November 2026.

## 01 SOURCE REGISTER

**[S1] EduHub - Online Learning Platform, project specification (provided PDF).**
user-provided attachment

**[R1] Microsoft - .NET releases and support**
https://learn.microsoft.com/en-us/dotnet/core/releases-and-support

**[R2] Microsoft - ASP.NET Core MVC overview**
https://learn.microsoft.com/en-us/aspnet/core/mvc/overview?view=aspnetcore-10.0

**[R3] Microsoft - EF Core SQL Server provider**
https://learn.microsoft.com/en-us/ef/core/providers/sql-server/

**[R4] Microsoft - Azure user delegation SAS with .NET**
https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blob-user-delegation-sas-create-dotnet

**[R5] Stripe - Checkout fulfillment / webhooks**
https://docs.stripe.com/checkout/fulfillment

**[R6] iText - iTextSharp EOL and licensing**
https://www.itextpdf.com/products/itextsharp

**[R7] Microsoft - Front Door & classic CDN migration FAQ**
https://learn.microsoft.com/en-us/azure/frontdoor/migration-faq

**[R8] Microsoft - Azure App Service ASP.NET quickstart**
https://learn.microsoft.com/en-us/azure/app-service/quickstart-dotnetcore

**[R9] Video.js - HTML installation guide**
https://videojs.org/docs/guides/installation/html

**[R10] Bootstrap - progress UI guidance**
https://getbootstrap.com/docs/5.3/components/progress/

**[R11] Chart.js - documentation**
https://www.chartjs.org/docs/latest/

**[R12] Microsoft - ASP.NET Core Identity**
https://learn.microsoft.com/en-us/aspnet/core/security/authentication/identity